

MONESH S

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PROFESSIONAL SUMMARY

Assistant Professor and AI Researcher specializing in Computer Vision, Scene Graph Generation, Graph Neural Networks, Vision Transformers, and Deep Learning. Currently pursuing doctoral research at Vellore Institute of Technology with a focus on graph-based visual reasoning and multimodal scene understanding. Published research in reputed international journals and IEEE conferences, and filed multiple Indian patents. Passionate about delivering quality engineering education while advancing research in intelligent visual perception, graph learning, and explainable artificial intelligence.

RESEARCH INTERESTS

- Computer Vision
 - Scene Graph Generation
 - Graph Neural Networks
 - Vision Transformers
 - Deep Learning
 - Visual Relationship Detection
 - Vision-Language Models
 - Multimodal Learning
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EDUCATION

Doctor of Philosophy (Ph.D.) (*Pursuing*)

Vellore Institute of Technology (VIT), Vellore

Research Area: Computer Vision, Scene Graph Generation and Graph Learning

Expected Completion: 2026

Master of Engineering (M.E.)

Computer Science (Specialization: Big Data Analytics)

College of Engineering Guindy (Anna University), Chennai

2019 – 2021

Bachelor of Technology (B.Tech.)

Information Technology

Hindustan Institute of Technology and Science, Chennai

2015 – 2019

ACADEMIC EXPERIENCE

Assistant Professor (Junior)

Vellore Institute of Technology (VIT), Vellore

August 2025 – July 2026

- Teach undergraduate and postgraduate courses in Artificial Intelligence, Machine Learning, Computer Vision, Predictive Data Analytics, Financial Data Analytics, Python Programming, and Design & Analysis of Algorithms.
 - Mentor undergraduate and postgraduate students in academic projects and research activities.
 - Conduct research in Scene Graph Generation, Graph Neural Networks, Vision Transformers, and Deep Learning.
 - Publish research articles in peer-reviewed international journals and IEEE conferences.
 - Review manuscripts for reputed international journals in Artificial Intelligence and Computer Vision.
 - Participate in curriculum development, outcome-based education, and academic assessment activities.
 - Collaborate with interdisciplinary research groups in AI and Computer Vision.
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Research Scholar

Vellore Institute of Technology (VIT), Vellore

August 2022 – July 2025

- Conducted doctoral research in Computer Vision and Graph Learning.
 - Developed transformer-based architectures for Scene Graph Generation and visual relationship prediction.
 - Designed graph-based reasoning frameworks for intelligent scene understanding.
 - Published research articles in international journals and conferences.
 - Filed multiple Indian patents in Artificial Intelligence and Computer Vision.
 - Participated in collaborative interdisciplinary research projects.
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COURSES TAUGHT

- Artificial Intelligence
 - Machine Learning
 - Computer Vision
 - Predictive Data Analytics
 - Financial Data Analytics
 - Design and Analysis of Algorithms
 - Python Programming
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CURRENT RESEARCH PROJECTS

1. Dynamic Scene Graph Intelligence for Proactive Vehicle Safety (Ongoing)
 2. Space Scene Graph Intelligence for Autonomous Planetary Exploration (Concept Development)
 3. AI-Driven Knowledge Discovery from Boogar's Literature (Future Idea)
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SELECTED PUBLICATIONS

Journal Publications

1. Monesh, S., & Senthilkumar, N. C. BiFormer for Scene Graph Generation Based on VisionNet with Taylor Hiking Optimization Algorithm. *IEEE Access*, 2025.

2. Senthilkumar, N. C., & Monesh, S. Review on Scene Graph Generation Methods. *Multiagent and Grid Systems*, 20(2), 2024.
 3. Monesh, S. Visual Relationship Prediction and Scene Graph Generation using Bi-GRU. *International Journal for Advanced Research, Ideas and Innovations in Technology (IJARIIT)*, 2021.
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Conference Publications

1. Monesh, S., & Senthilkumar, N. C. MATAN: Mixed Attention-Based Transformer Network with Visformer Features for Visual Relationship Modeling. *IEEE International Research Conference on Smart Computing and Systems Engineering (SCSE 2026)*, University of Kelaniya, Sri Lanka.
 2. Sharmila, V. C., Monesh, S., Aayush, R., Karesh, G., & Ibrahim, I. Digitized Bus Ticketing Framework. *1st International Conference on Innovations in Information and Communication Technology*, 2019.
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PUBLISHED PATENTS

1. Privacy-Preserving Distributed Surveillance System for Real-Time Suspicious Activity Detection Using Scene Graphs and Federated Learning (2026).
 2. Physics-Constrained Spatio-Temporal Scene Graph Validator for Video Authenticity Detection (2026).
 3. AI-Powered Smart Eyeglass Generates Scene Graph Knowledge for Clinical Surveillance (2025).
 4. AI-Powered Smartwatch for Food Intake Monitoring and Recommender System (2024).
 5. AI-Powered Tire Pressure Monitoring System for Heavy Vehicles (2024).
 6. Deep Learning-Based Automated Cow Distraction Device from Plastic Materials (2022).
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AWARDS & HONORS

- Raman Research Award (2025) for publishing research in an SCI-indexed journal.
- Dr. A.P.J. Abdul Kalam Award (2024) for publishing a review article in a Scopus-indexed journal.
- Presented the research paper "MATAN: Mixed Attention-Based Transformer Network with Visformer Features for Visual Relationship Modeling" at the 9th IEEE

PROFESSIONAL DEVELOPMENT

- Five-Day Faculty Development Program on Comprehensive Hands-on Sessions on Computer Vision and Artificial Intelligence Towards Applied Research, Teaching Learning Centre of Excellence (TLCE), VIT.
 - Five-Day Faculty Development Program on Deep Learning Architecture Concepts and GAN with Deepfake Techniques, Academic Staff College, VIT.
 - Faculty Development Program on Digital Twin and Deep Learning – AI-Driven Approaches for Smart Factories, Academic Staff College, VIT.
 - Faculty Development Program on Application of Optimization Techniques in Engineering, Academic Staff College, VIT.
 - Faculty Development Program on Research Opportunities after Ph.D., Academic Staff College, VIT.
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CERTIFICATIONS

- **NPTEL:** Computer Vision, IIT Kharagpur.
 - **Google Cloud:** Data Engineering, Big Data and Machine Learning on Google Cloud Platform Specialization.
 - **Google Cloud:** Feature Engineering on Google Cloud Platform.
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TECHNICAL SKILLS

Programming Languages

Python, C, C++, Java

Artificial Intelligence

Machine Learning, Deep Learning, Computer Vision, Scene Graph Generation, Graph Neural Networks, Vision Transformers, Explainable AI

Frameworks & Libraries

TensorFlow, PyTorch, Keras, OpenCV, Scikit-learn

Research Tools

LaTeX, Jupyter Notebook, Google Colab

PROFESSIONAL MEMBERSHIPS

- IEEE (Member ID: 95177799)

LANGUAGES

- English
- Tamil
- Telugu (Conversational)

Signature

Monesh S